WORKSHEET

Chapter 8: Simple differential equations

1 Find the general solutions to the following differential equations.

a $\frac{dy}{dx} = 4x^3 - 3x^2 + 8x - 1$

b $\frac{dy}{dx} = 3\sin(5x)$

c $\frac{dy}{dx} = 12x^4 + \frac{14}{x}$

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d $\frac{dy}{dx} = 8e^{-4x} - 6e^{3x}$

e $\frac{dy}{dx} = (1 - 2x)^2$

f $\frac{dy}{dx} = \frac{1}{(5 + 3x)^3}$

2 Find the general solutions to the following differential equations.

a $\frac{dy}{dx} = x\left(x^2 - 1\right)^{\frac{2}{3}}$

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b $\frac{dy}{dx} = 2\sec^2(3x)$

c $\frac{dx}{dy} = \frac{x^3 + 5x^2 - 4}{x^2}$

d $\frac{dx}{dy} = \frac{8x^2}{(x^3 + 2)^3}$

e $\frac{dy}{dx} = 3x\sqrt{1 - 2x^2}$

f $\frac{dy}{dx} = \frac{1}{\sqrt{25 - 16x^2}}$

3 Find the solutions to the following differential equations.

a $\frac{dy}{dx} = (x + 2)^{\frac{3}{2}}$ when $y(-2) = 3$

b $\frac{dy}{dx} = \frac{x^2 + 2x}{(x + 1)^2}$ when $y(2) = \frac{-2}{3}$

c $\frac{dy}{dx} = \frac{x}{(x^2 - 3)^3}$ when $y(2) = \frac{3}{4}$

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d $\frac{dy}{dx} = \frac{1}{4x^2+9}$ when $y(3) = 0$

e $\frac{dy}{dx} = \text{cosec}^2(x)$ when $y\left(\frac{\pi}{4}\right) = 5$

f $\frac{dy}{dx} = \frac{1}{\sqrt{3-x^2}}$ when $y\left(\frac{3}{2}\right) = \pi$

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4 Find the solutions to the following differential equations.

a $\frac{dx}{dy} = \frac{1}{\sqrt[3]{x^2}}$ when $y = 1, x = 1$

b $\frac{dx}{dy} = \frac{x^2}{\sqrt{1 - x^6}}$ when $y = \pi, x = 1$

c $\frac{dy}{dx} = \frac{1}{\sqrt{4 - x^2}}$ when $y = 0, x = 1$

d $\frac{dy}{dx} = \frac{3}{1 + 4x^2}$ when $y = \pi, x = \frac{1}{2}$

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e $\frac{dy}{dx} = 20x\left(2x^2 + 1\right)^4$ when $y(0) = -2$

f $\frac{dy}{dx} = \sin(x)\cos^2(x)$ when $y(0) = 0$

Answers

1 **a** $x^4 - x^3 + 4x^2 - x + c$

b $\dfrac{-3\cos(5x)}{5} + c$

c $\dfrac{12x^5}{5} + 14\ln(x) + c$

d $-2e^{3x} - 2e^{-4x} + c$

e $x - 2x^2 + \dfrac{4x^3}{3} + c$

f $\dfrac{-1}{6(3x+5)^2} + c$

2 **a** $\dfrac{3(x^2-1)^{\frac{5}{3}}}{10} + c$

b $\dfrac{2\tan(3x)}{3} + c$

c $\dfrac{x^2}{2} + 5x + \dfrac{4}{x} + c$

d $\dfrac{-4}{3(x^3+2)^2} + c$

e $\dfrac{-(1-2x^2)^{\frac{3}{2}}}{2} + c$

f $\dfrac{1}{4}\arcsin\left(\dfrac{4x}{5}\right) + c$

3 **a** $\dfrac{2}{5}(x+2)^{\frac{5}{2}} + 3$

b $\dfrac{x^2}{x+1} - 2$

c $-\dfrac{1}{4(x^2-3)^2} + 1$

d $\dfrac{1}{3}\arctan\left(\dfrac{x}{3}\right) - \dfrac{\pi}{12}$

e $-\dfrac{1}{2}\cot(2x) + 5$

f $\dfrac{1}{2}\arcsin\left(\dfrac{x\sqrt{3}}{3}\right) + \dfrac{2\pi}{3}$

4 **a** $3\sqrt[3]{x} - 2$

b $\dfrac{1}{3}\arcsin(x^3) + \dfrac{5\pi}{6}$

c $\arcsin\left(\dfrac{x}{2}\right) - \dfrac{\pi}{6}$

d $\dfrac{3}{2}\arctan(2x) + \dfrac{5\pi}{8}$

e $(2x^2+1)^5 - 3$

f $\dfrac{-\cos^3(x)}{3} + \dfrac{1}{3}$

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